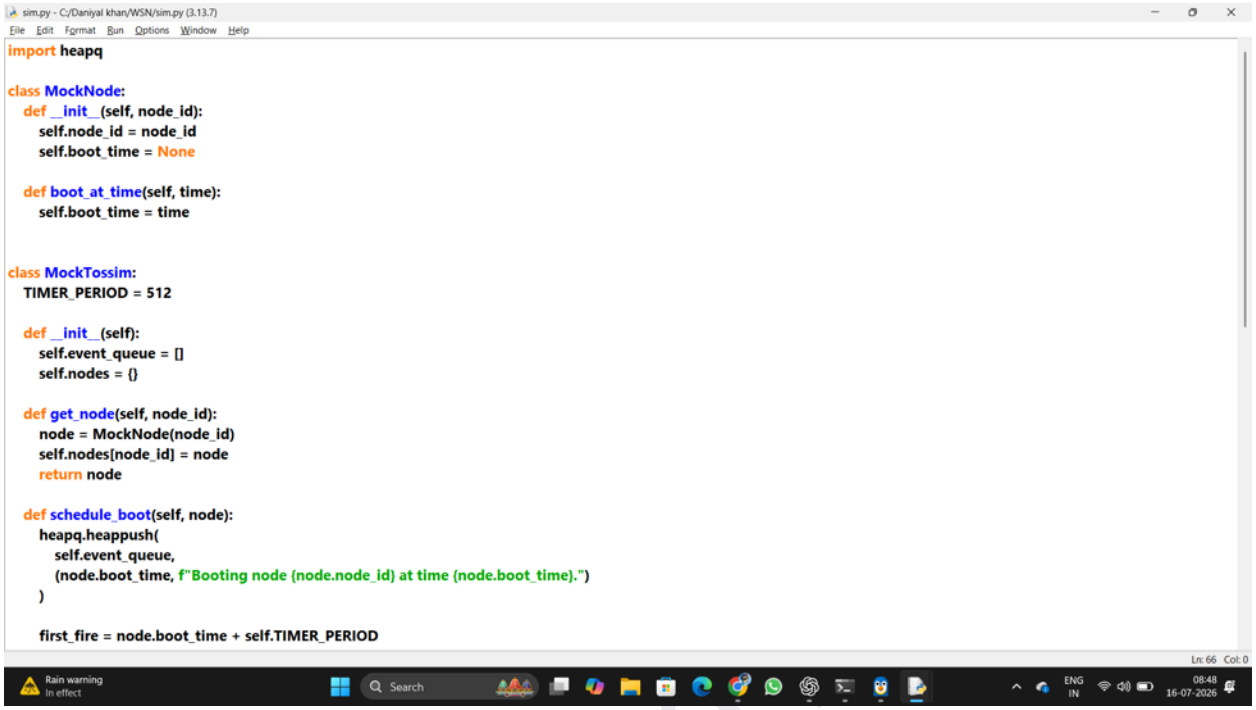
SHETH LUJ COLLEGE

WSN PRACTICAL - 3



```
import heapq

class MockNode:
    def __init__(self, node_id):
        self.node_id = node_id
        self.boot_time = None

    def boot_at_time(self, time):
        self.boot_time = time

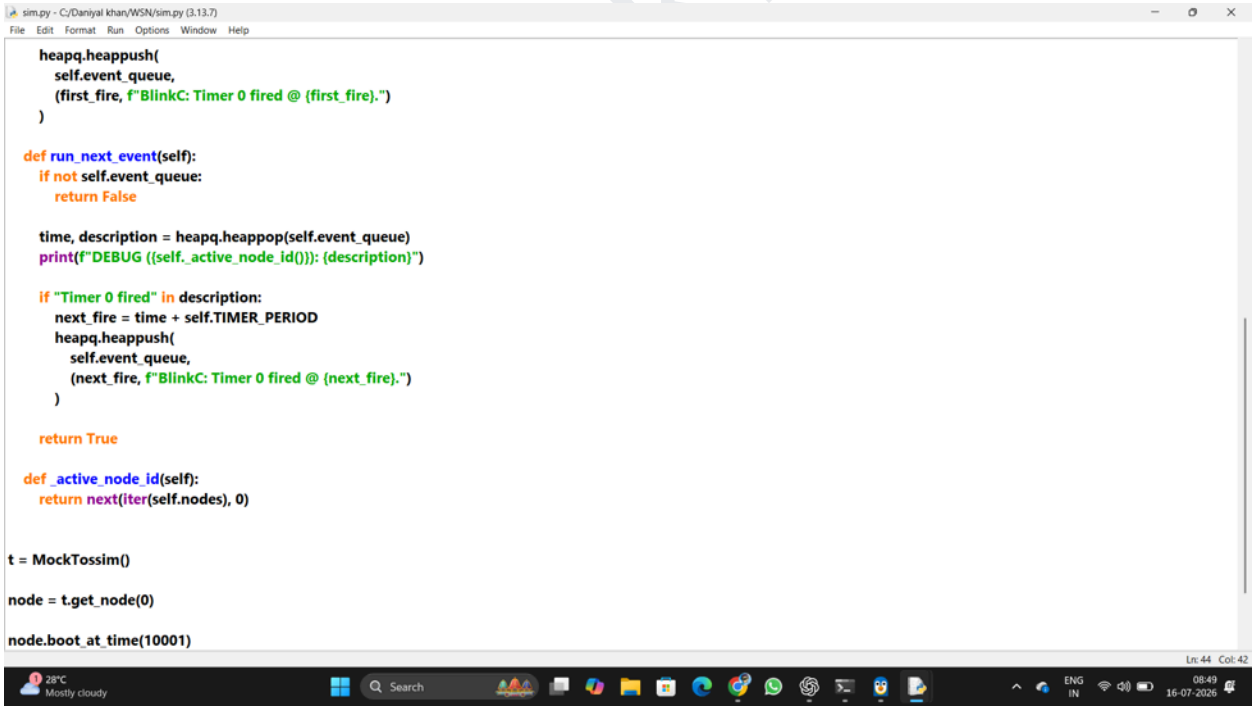

class MockTossim:
    TIMER_PERIOD = 512

    def __init__(self):
        self.event_queue = []
        self.nodes = {}

    def get_node(self, node_id):
        node = MockNode(node_id)
        self.nodes[node_id] = node
        return node

    def schedule_boot(self, node):
        heapq.heappush(
            self.event_queue,
            (node.boot_time, f"Booting node {node.node_id} at time {node.boot_time}.")
        )

        first_fire = node.boot_time + self.TIMER_PERIOD
```



```
        heapq.heappush(
            self.event_queue,
            (first_fire, f"BlinkC: Timer 0 fired @ {first_fire}.")
        )

    def run_next_event(self):
        if not self.event_queue:
            return False

        time, description = heapq.heappop(self.event_queue)
        print(f"DEBUG ({self._active_node_id()}): {description}")

        if "Timer 0 fired" in description:
            next_fire = time + self.TIMER_PERIOD
            heapq.heappush(
                self.event_queue,
                (next_fire, f"BlinkC: Timer 0 fired @ {next_fire}.")
            )

        return True

    def _active_node_id(self):
        return next(iter(self.nodes), 0)


t = MockTossim()

node = t.get_node(0)

node.boot_at_time(10001)
```

TIWARI ANKITA
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WSN PRACTICAL - 3

```
sim.py - C:/Daniyal khan/WSN/sim.py (3.13.7)
File  Edit  Format  Run  Options  Window  Help
  def run_next_event(self):
    if not self.event_queue:
      return False

    time, description = heapq.heappop(self.event_queue)
    print(f"DEBUG ({self._active_node_id()}): {description}")

    if "Timer 0 fired" in description:
      next_fire = time + self.TIMER_PERIOD
      heapq.heappush(
        self.event_queue,
        (next_fire, f"BlinkC: Timer 0 fired @ {next_fire}.")
      )

    return True

  def _active_node_id(self):
    return next(iter(self.nodes), 0)


t = MockTossim()

node = t.get_node(0)

node.boot_at_time(10001)
t.schedule_boot(node)

for i in range(10):
  t.run_next_event()
```

Output/;-

```
*IDLE Shell 3.13.7*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Daniyal khan/WSN/sim.py ==========
DEBUG (0): Booting node 0 at time 10001.
DEBUG (0): BlinkC: Timer 0 fired @ 10513.
DEBUG (0): BlinkC: Timer 0 fired @ 11025.
DEBUG (0): BlinkC: Timer 0 fired @ 11537.
DEBUG (0): BlinkC: Timer 0 fired @ 12049.
DEBUG (0): BlinkC: Timer 0 fired @ 12561.
DEBUG (0): BlinkC: Timer 0 fired @ 13073.
DEBUG (0): BlinkC: Timer 0 fired @ 13585.
DEBUG (0): BlinkC: Timer 0 fired @ 14097.
DEBUG (0): BlinkC: Timer 0 fired @ 14609.
>>>
```

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